

Neural networks

Weekend 1, Friday afternoon

Carlos Cotrini

Friday 4 September 2026

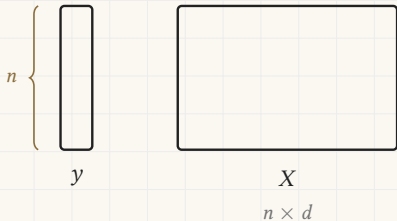
1

Section 1

Linear regression in one picture

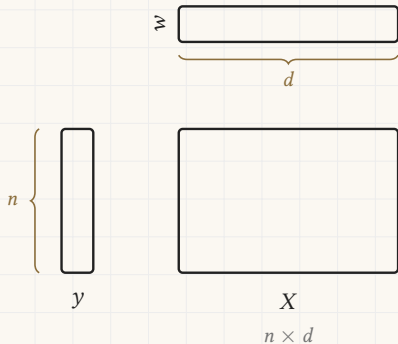
Linear regression with gradient descent

The whole of this morning, drawn once. Each box is a piece of data, and its shape is the shape of the data.



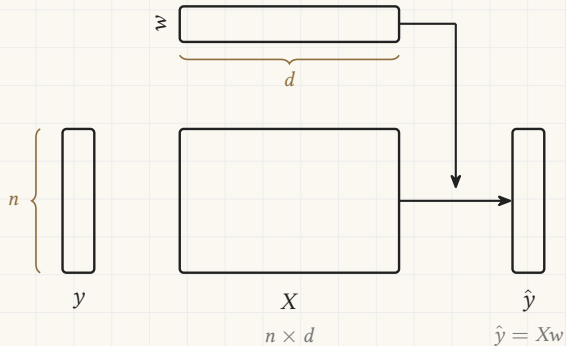
Linear regression with gradient descent

The whole of this morning, drawn once. Each box is a piece of data, and its shape is the shape of the data.



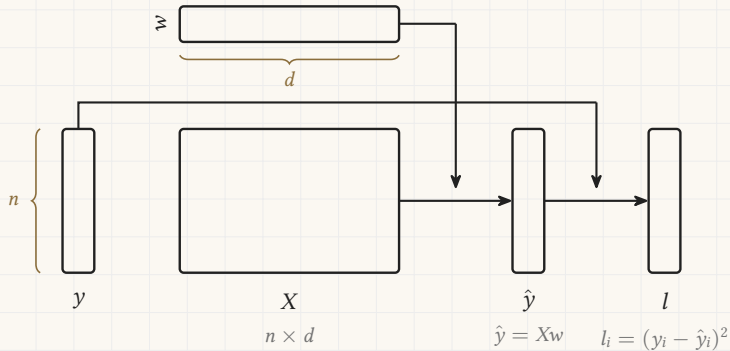
Linear regression with gradient descent

The whole of this morning, drawn once. Each box is a piece of data, and its shape is the shape of the data.



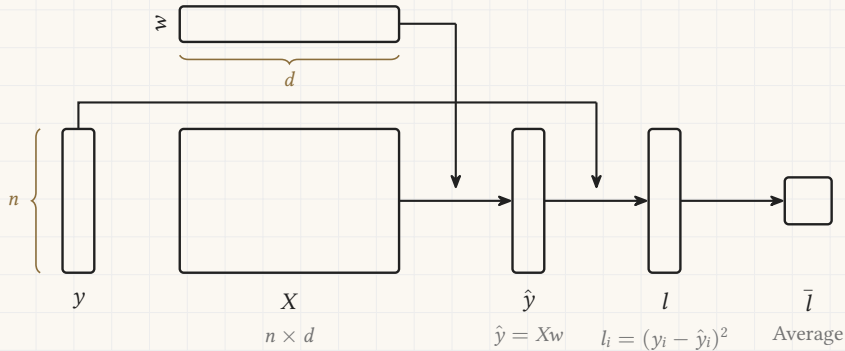
Linear regression with gradient descent

The whole of this morning, drawn once. Each box is a piece of data, and its shape is the shape of the data.



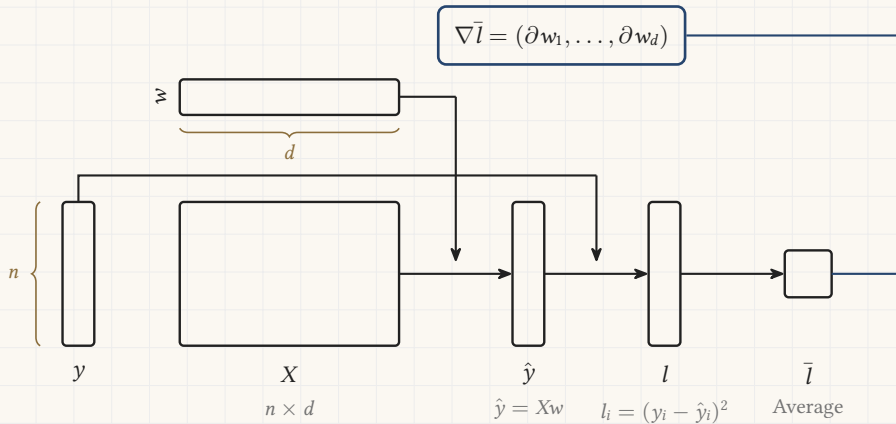
Linear regression with gradient descent

The whole of this morning, drawn once. Each box is a piece of data, and its shape is the shape of the data.



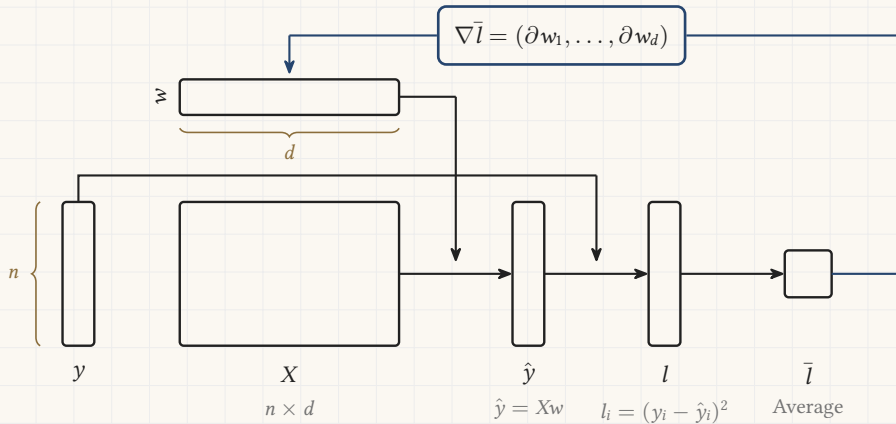
Linear regression with gradient descent

The whole of this morning, drawn once. Each box is a piece of data, and its shape is the shape of the data.



Linear regression with gradient descent

The whole of this morning, drawn once. Each box is a piece of data, and its shape is the shape of the data.



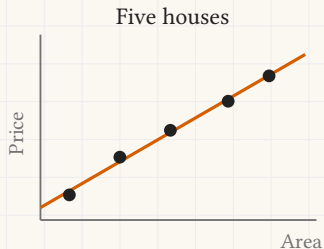
2

Section 2

What a linear model cannot fit

A linear model does not fit everything

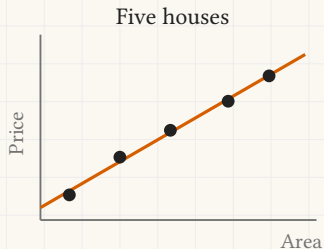
Three sets of measurements, and the best straight line through each.



The largest miss is 83'413 CHF,
which is 3.8 percent of the
price range.

A linear model does not fit everything

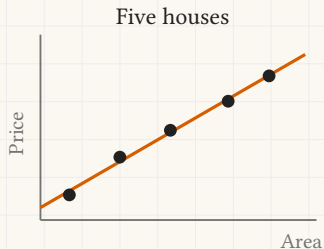
Three sets of measurements, and the best straight line through each.



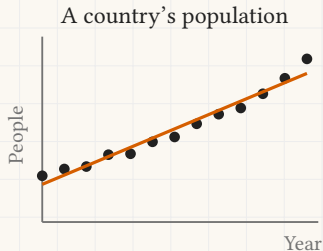
The largest miss is 83'413 CHF,
which is 3.8 percent of the
price range.

A linear model does not fit everything

Three sets of measurements, and the best straight line through each.



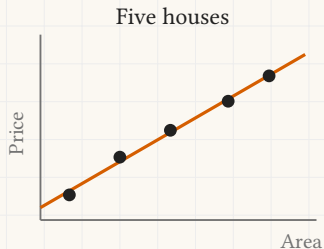
The largest miss is 83'413 CHF, which is 3.8 percent of the price range.



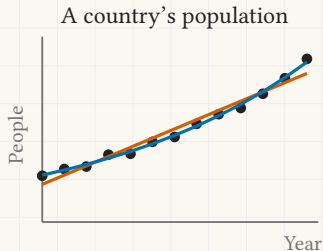
The straight line misses by 12.4 percent of the range. The curve $y = e^{wx}$ misses by 3.2 percent. Synthetic data.

A linear model does not fit everything

Three sets of measurements, and the best straight line through each.



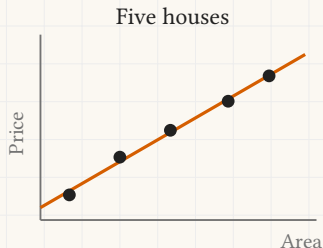
The largest miss is 83'413 CHF, which is 3.8 percent of the price range.



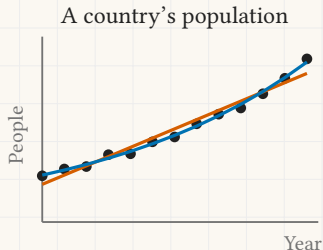
The straight line misses by 12.4 percent of the range. The curve $y = e^{wx}$ misses by 3.2 percent. Synthetic data.

A linear model does not fit everything

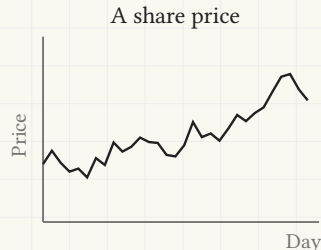
Three sets of measurements, and the best straight line through each.



The largest miss is 83'413 CHF, which is 3.8 percent of the price range.

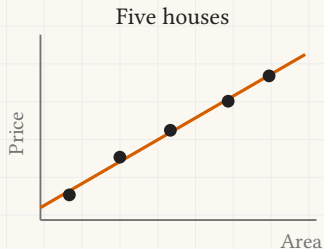


The straight line misses by 12.4 percent of the range. The curve $y = e^{wx}$ misses by 3.2 percent. Synthetic data.

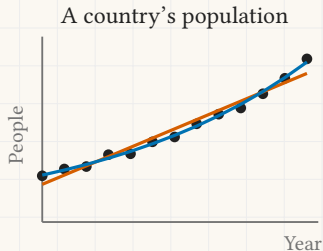


A linear model does not fit everything

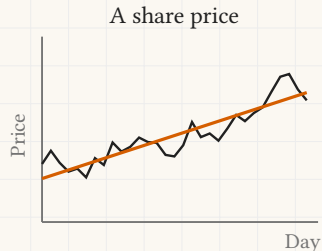
Three sets of measurements, and the best straight line through each.



The largest miss is 83'413 CHF, which is 3.8 percent of the price range.



The straight line misses by 12.4 percent of the range. The curve $y = e^{wx}$ misses by 3.2 percent. Synthetic data.



The straight line misses by 32.8 percent of the range, and no elementary curve does better. Synthetic data.

3

Section 3

Universal approximators

Universal approximators

Let K be a closed and bounded set of inputs, and let $\mathcal{F}$ be a family of functions from K to the real numbers.

$\mathcal{F}$ is a **universal approximator** on K if for every continuous $g : K \rightarrow \mathbb{R}$ and every $\varepsilon > 0$ there is an $f \in \mathcal{F}$ with $\max_{x \in K} |g(x) - f(x)| < \varepsilon$.

Universal approximators

Let K be a closed and bounded set of inputs, and let $\mathcal{F}$ be a family of functions from K to the real numbers.

$\mathcal{F}$ is a **universal approximator** on K if for every continuous $g : K \rightarrow \mathbb{R}$ and every $\varepsilon > 0$ there is an $f \in \mathcal{F}$ with $\max_{x \in K} |g(x) - f(x)| < \varepsilon$.

Two families are known to meet this definition.

Universal approximators

Let K be a closed and bounded set of inputs, and let $\mathcal{F}$ be a family of functions from K to the real numbers.

$\mathcal{F}$ is a **universal approximator** on K if for every continuous $g : K \rightarrow \mathbb{R}$ and every $\varepsilon > 0$ there is an $f \in \mathcal{F}$ with $\max_{x \in K} |g(x) - f(x)| < \varepsilon$.

Two families are known to meet this definition.

- 1 The polynomials, on a closed interval. That is the Weierstrass approximation theorem, and it dates from 1885.

Universal approximators

Let K be a closed and bounded set of inputs, and let $\mathcal{F}$ be a family of functions from K to the real numbers.

$\mathcal{F}$ is a **universal approximator** on K if for every continuous $g : K \rightarrow \mathbb{R}$ and every $\varepsilon > 0$ there is an $f \in \mathcal{F}$ with $\max_{x \in K} |g(x) - f(x)| < \varepsilon$.

Two families are known to meet this definition.

- 1 The polynomials, on a closed interval. That is the Weierstrass approximation theorem, and it dates from 1885.
- 2 The neural networks. The rest of this lecture builds one, and the last slide states the theorem.

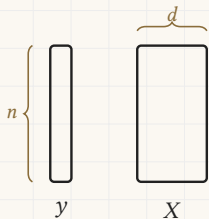
4

Section 4

Neural networks

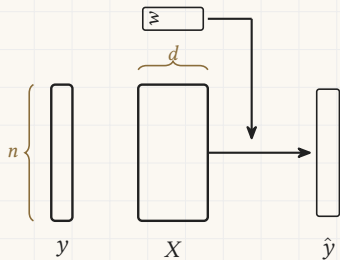
From one weight vector to a layer

The same X , and more than one weight vector. Every block is drawn to scale.



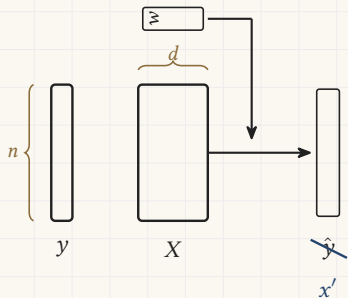
From one weight vector to a layer

The same X , and more than one weight vector. Every block is drawn to scale.



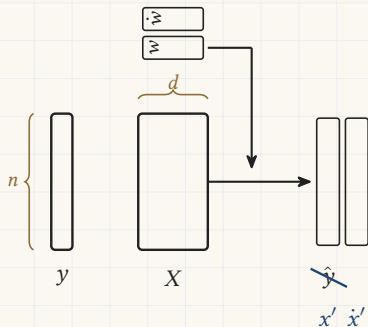
From one weight vector to a layer

The same X , and more than one weight vector. Every block is drawn to scale.



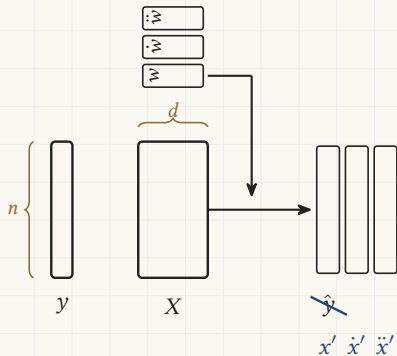
From one weight vector to a layer

The same X , and more than one weight vector. Every block is drawn to scale.



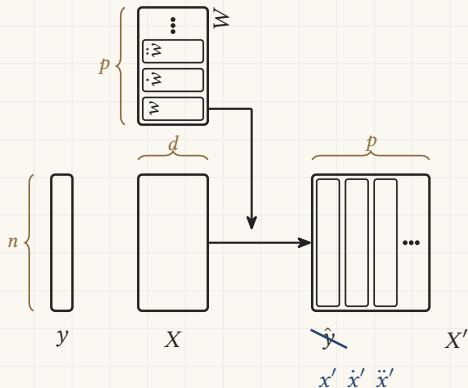
From one weight vector to a layer

The same X , and more than one weight vector. Every block is drawn to scale.



From one weight vector to a layer

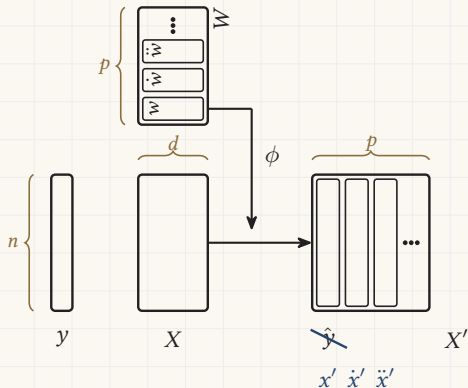
The same X , and more than one weight vector. Every block is drawn to scale.



W is laid on its side, so one of its columns is a row up there. Each row makes one column of X' , and the two are drawn at the same pitch, so you can count them off against each other. W has d rows, which is what makes the product defined, and p columns, which is a number you choose.

From one weight vector to a layer

The same X , and more than one weight vector. Every block is drawn to scale.

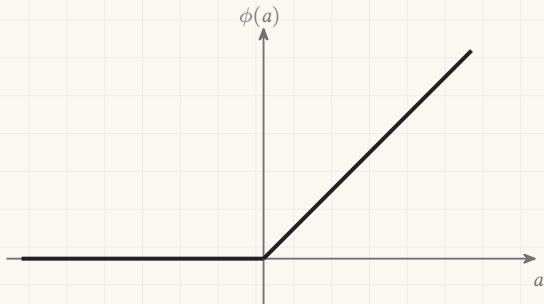


$$X' = \phi(XW)$$

W is laid on its side, so one of its columns is a row up there. Each row makes one column of X' , and the two are drawn at the same pitch, so you can count them off against each other. W has d rows, which is what makes the product defined, and p columns, which is a number you choose.

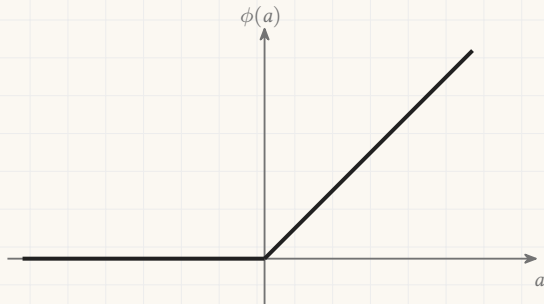
The activation function

ϕ is applied to every entry on its own, and it is the only piece of the network that is not a matrix product.



The activation function

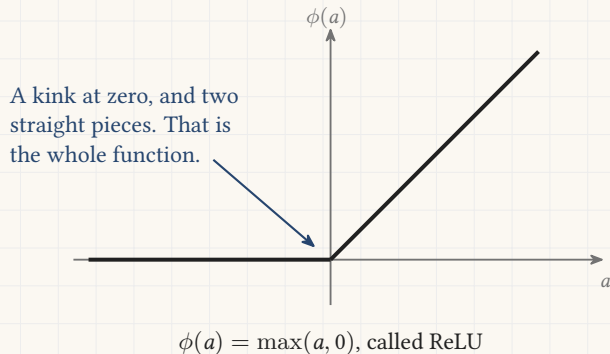
ϕ is applied to every entry on its own, and it is the only piece of the network that is not a matrix product.



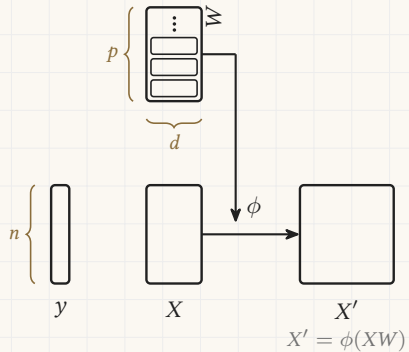
$$\phi(a) = \max(a, 0), \text{ called ReLU}$$

The activation function

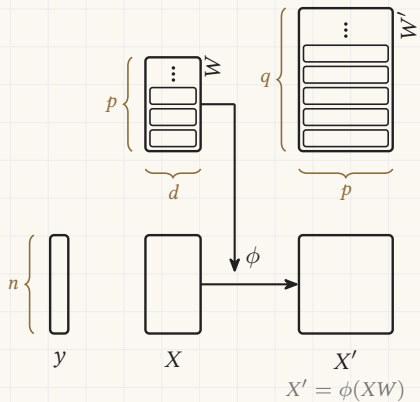
ϕ is applied to every entry on its own, and it is the only piece of the network that is not a matrix product.



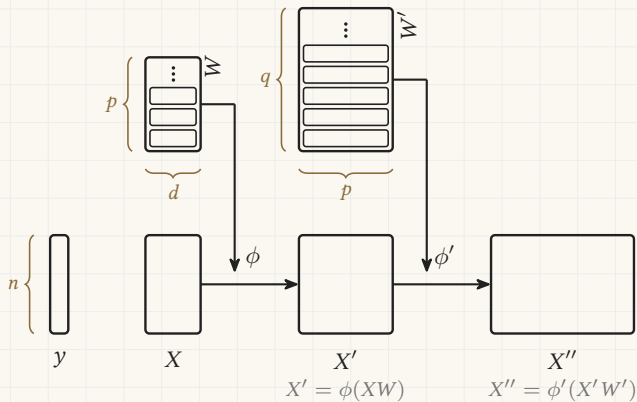
A neural network



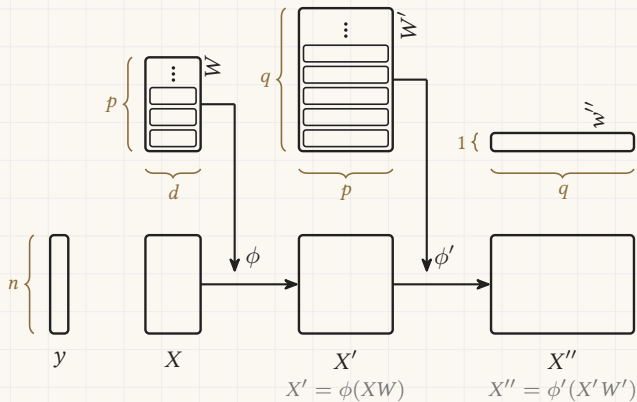
A neural network



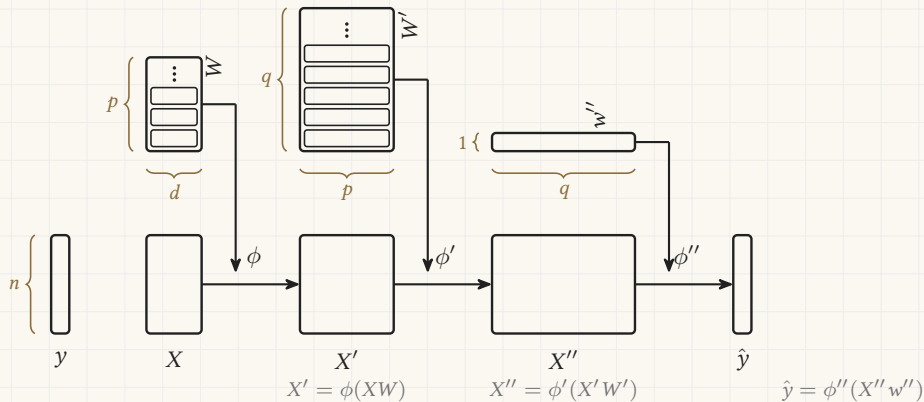
A neural network



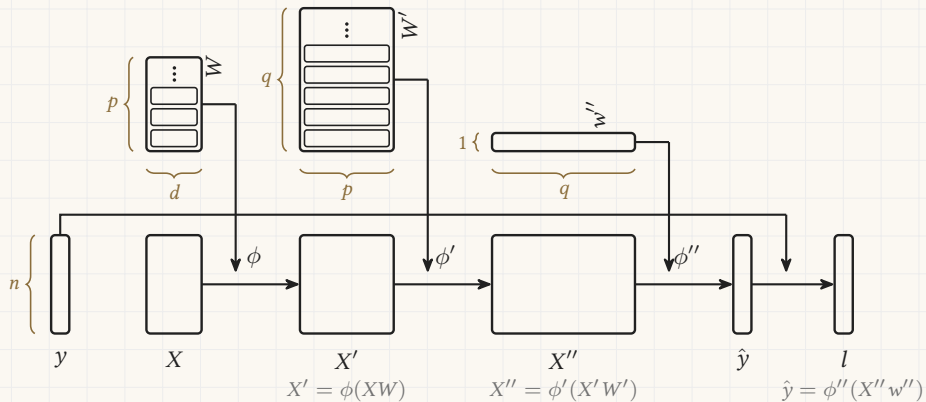
A neural network



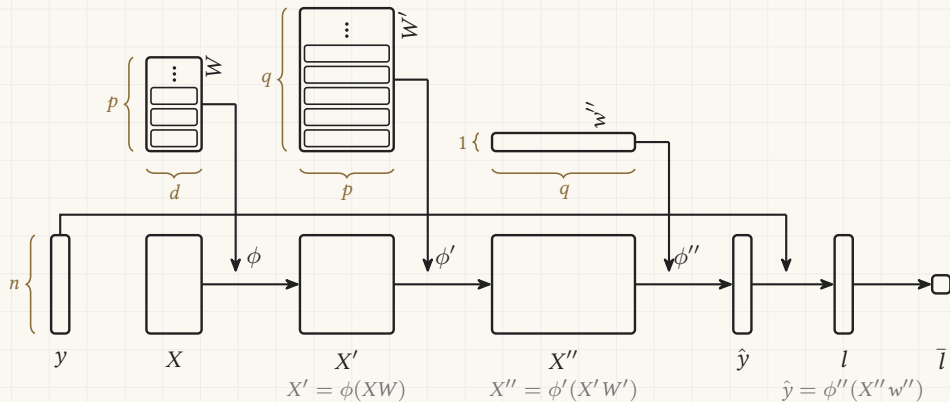
A neural network



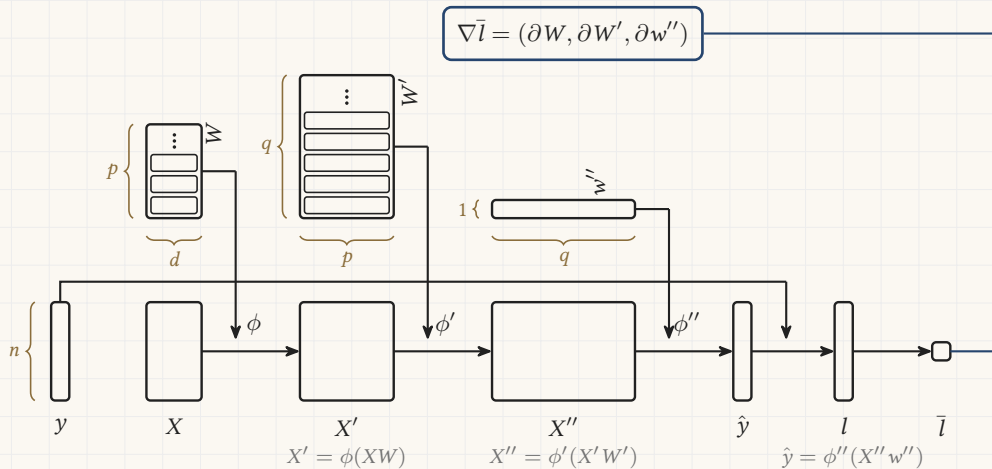
A neural network



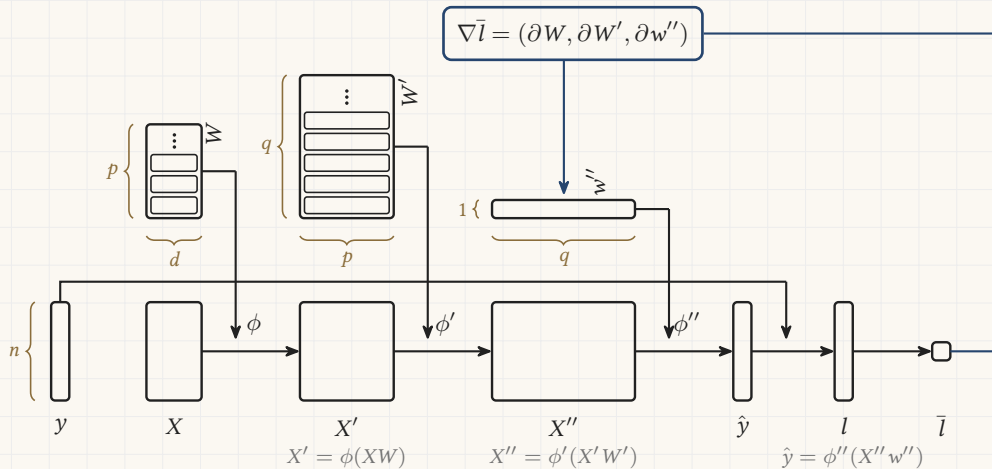
A neural network



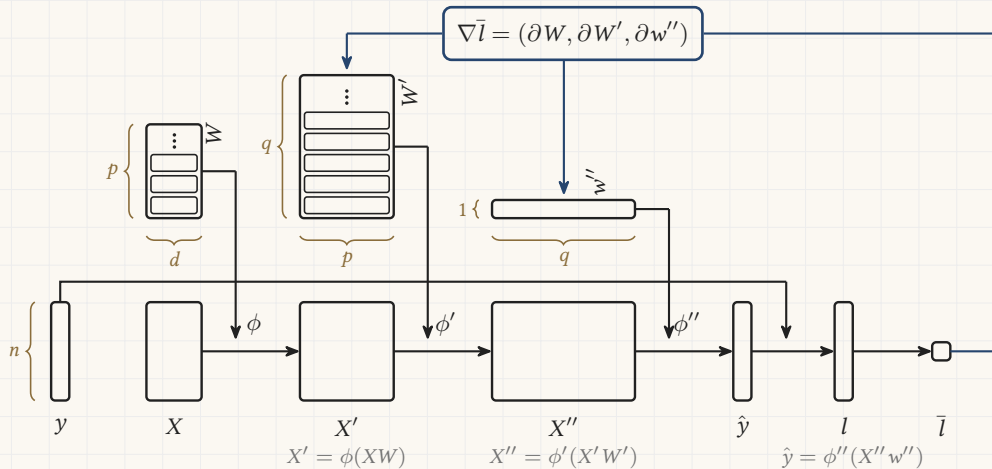
A neural network



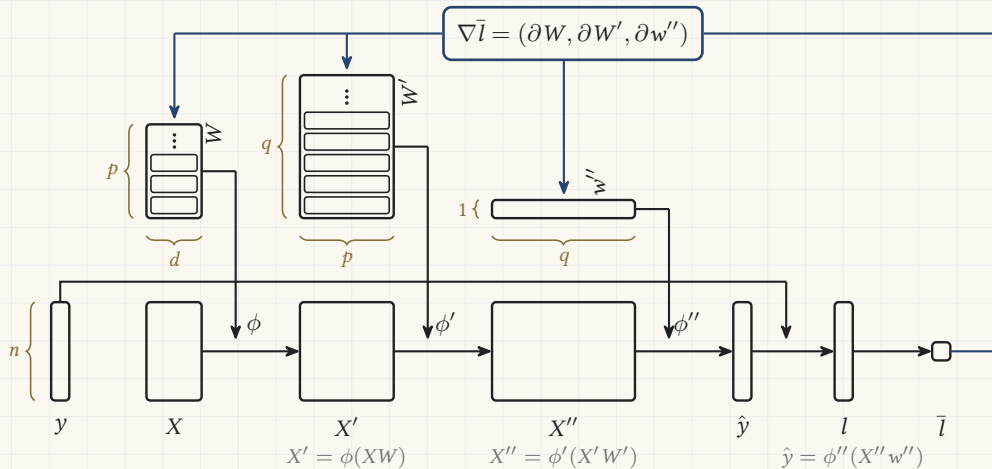
A neural network



A neural network



A neural network



The same network, drawn as neurons

One circle for every column of every weight block.

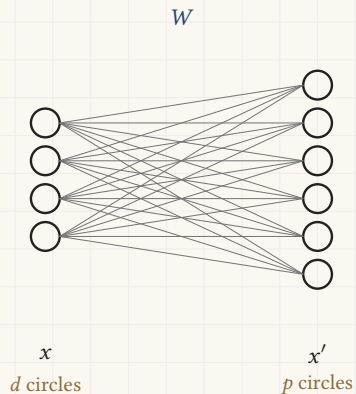


x

d circles

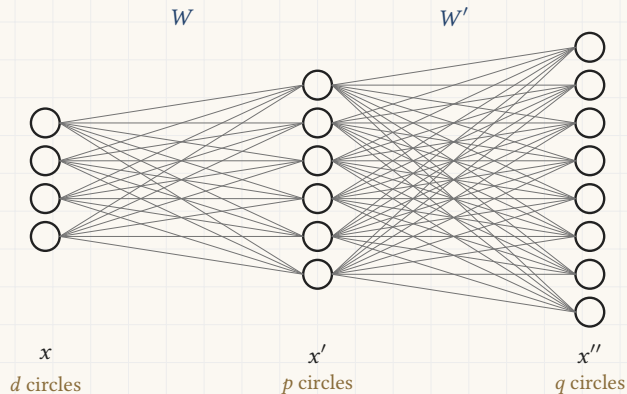
The same network, drawn as neurons

One circle for every column of every weight block.



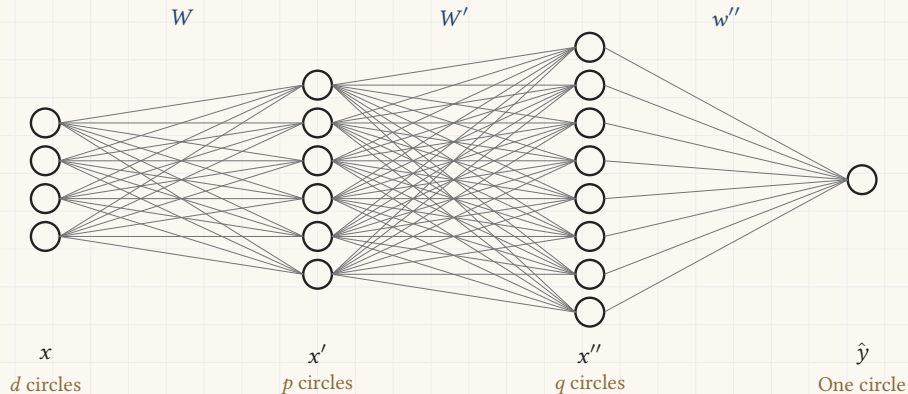
The same network, drawn as neurons

One circle for every column of every weight block.



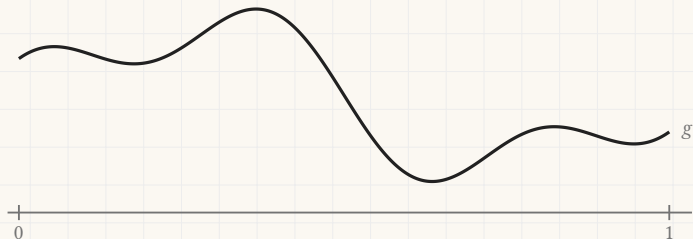
The same network, drawn as neurons

One circle for every column of every weight block.



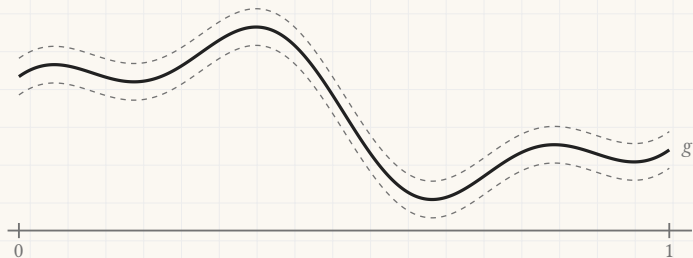
The universal approximation theorem

A continuous target, and networks with p hidden units drawn over it. The band is the accuracy $\varepsilon = 0.25$.



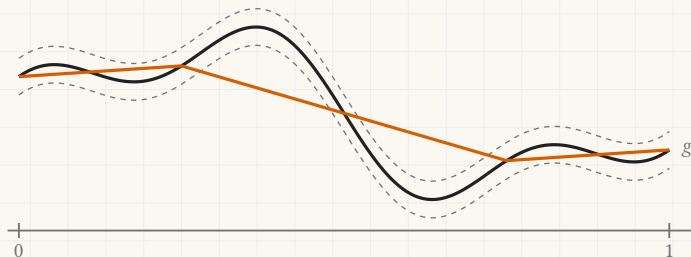
The universal approximation theorem

A continuous target, and networks with p hidden units drawn over it. The band is the accuracy $\varepsilon = 0.25$.



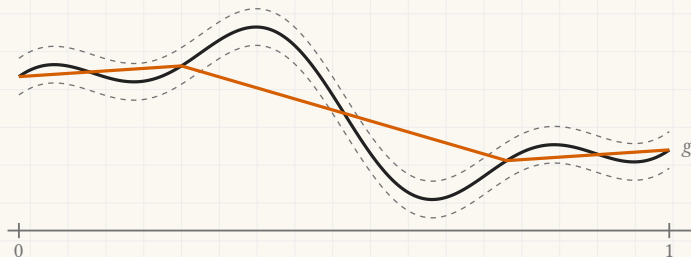
The universal approximation theorem

A continuous target, and networks with p hidden units drawn over it. The band is the accuracy $\varepsilon = 0.25$.



The universal approximation theorem

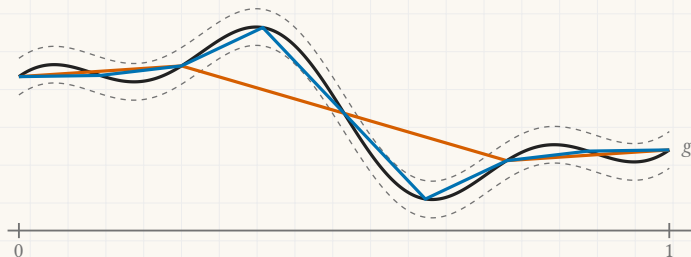
A continuous target, and networks with p hidden units drawn over it. The band is the accuracy $\varepsilon = 0.25$.



$p = 4$, largest miss 0.85043

The universal approximation theorem

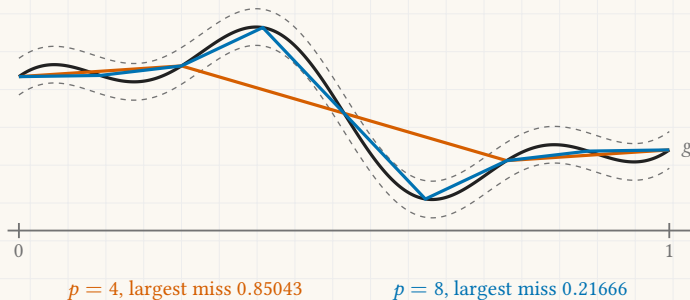
A continuous target, and networks with p hidden units drawn over it. The band is the accuracy $\varepsilon = 0.25$.



$p = 4$, largest miss 0.85043

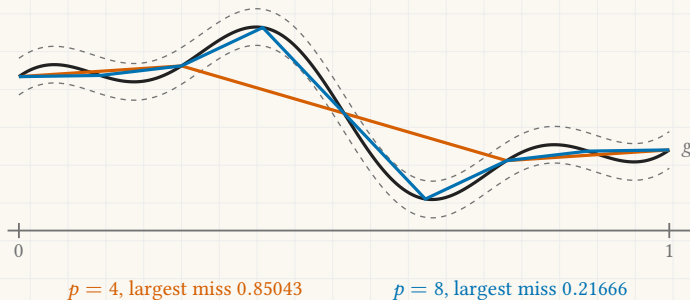
The universal approximation theorem

A continuous target, and networks with p hidden units drawn over it. The band is the accuracy $\varepsilon = 0.25$.



The universal approximation theorem

A continuous target, and networks with p hidden units drawn over it. The band is the accuracy $\varepsilon = 0.25$.



A network with one hidden layer, and enough units in it, is a universal approximator.

The largest miss falls to 0.06944 at $p = 16$, to 0.01788 at $p = 32$ and to 0.00448 at $p = 64$.

Cybenko, 1989. Hornik, Stinchcombe and White, 1989.